

CSC430/530 – Database Management Systems

Assignment #5 – Indexing Structures

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Consider a disk with a block size **B = 1024 bytes** and a block pointer **P = 12 bytes**. Suppose file has **r = 69632** EMPLOYEE records of fixed length.

Each record has following attributes:

- First_name (60 bytes);
- SSN (18 bytes);
- Dnumber (18 bytes);
- Home_address (80 bytes);
- Contact_phone (20 bytes);
- BDate (16 bytes);
- Gender (2 byte);
- Job_code (8 bytes);
- Salary (8 bytes);
- Additional 2 bytes are used as a deletion marker.

a. Assuming an unspanned organization, calculate following.

***To get full points, please, show all formulas and calculations.**

- Record size R (in bytes).

$$R = \sum \text{attributes} = 60 + 18 + 18 + 80 + 20 + 16 + 2 + 8 + 8 + 2 = 232$$

- Blocking factor bfr.

$$\text{bfr} = \lfloor B / R \rfloor = \lfloor 1024 / 232 \rfloor = 4$$

- Number of file blocks b.

$$b = \lceil r / \text{bfr} \rceil = \lceil 69632 / 4 \rceil = 17408$$

- b. Assuming file is ordered by key attribute **Ssn** and primary index is constructed on this key attribute, calculate following.

***To get full points, please, show all formulas and calculations.**

- Index blocking factor **bfr_i**.

$$R_i = \text{key} + \text{pointer} = 18 + 12 = 30$$

$$\text{bfr}_i = \lfloor B / R_i \rfloor = \lfloor 1024 / 30 \rfloor = 34$$

- Total number of indexes **r_i** and number of index blocks **b_i**.

$$r_i = b$$

b was calculated in question a. so **r_i = 17408**

$$b_i = \lceil r_i / \text{bfr}_i \rceil = \lceil 17408 / 34 \rceil = 512$$

- Total number of accesses needed to search for a record using primary index.

$$\text{bin search} = \lceil \log(b_i) \rceil = \lceil \log(512) \rceil = 9$$

$$\text{accesses} = \text{bin search} + 1 = 9 + 1 = 10$$